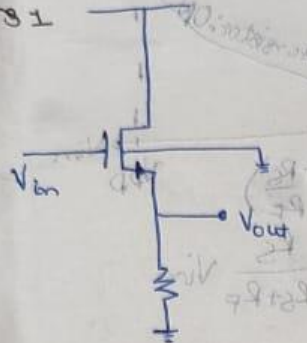


Assignment - 3

Mukul Groel
220108037

Ans 1



Given to us that transistor is biased in saturation

As we know that the input-output characteristics can be expressed as

$$\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{th} - V_{out})^2 R_S = V_{out}$$

* If we neglect the channel length modulation above.

For small-signal gain A_V -:

$$\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{th} - V_{out}) \left(1 - \frac{\partial V_{th}}{\partial V_{in}} - \frac{\partial V_{out}}{\partial V_{in}} \right) R_S = \frac{\partial V_{out}}{\partial V_{in}}$$

$$\left[\frac{\partial V_{th}}{\partial V_{in}} = \frac{\partial V_{th}}{\partial V_{SB}} \times \frac{\partial V_{SB}}{\partial V_{in}} = R \frac{\partial V_{out}}{\partial V_{in}} \right]$$

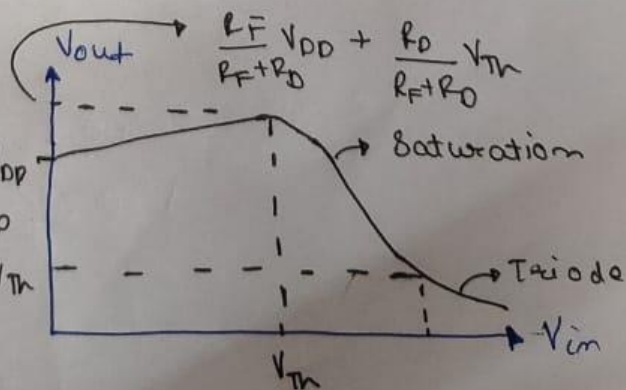
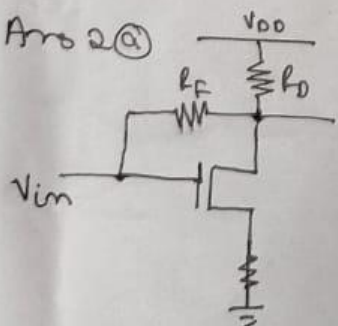
$$\Rightarrow \frac{\partial V_{out}}{\partial V_{in}} = \frac{\mu_n C_{ox} \frac{W}{L} (V_{in} - V_{th} - V_{out}) R_S}{1 + \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{th} - V_{out}) R_S (1 + R)}$$

And we know that

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{th} - V_{out})$$

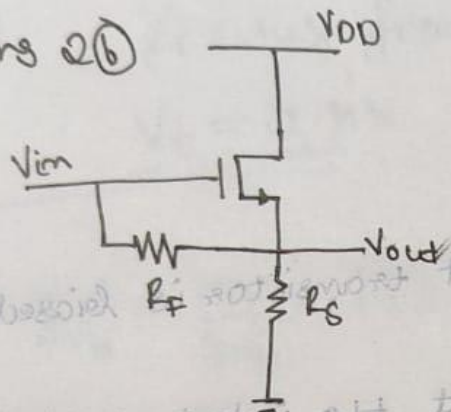
$$\Rightarrow A_V = \frac{g_m R_S}{1 + (g_m + g_{mb}) R_S}$$

Ans 2 @

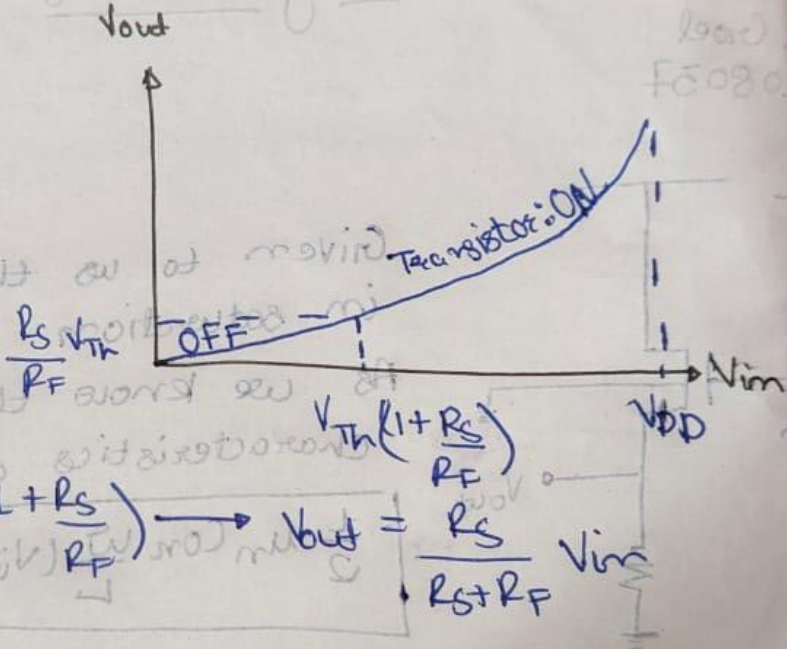


$$\text{If } V_{in} < V_{th} \rightarrow V_{out} = \frac{R_F}{R_F + R_D} V_{DD} + \frac{R_D}{R_F + R_D} V_{in}$$

Ans 2b)



5 - Transistor



If $V_{in} < V_{TN}(1 + \frac{R_S}{R_F}) \rightarrow V_{out} = 0$

If $V_{in} > V_{TN}(1 + \frac{R_S}{R_F}) \rightarrow V_{out} = \frac{R_S}{R_S + R_F} V_{in}$

the magnitude of the channel length modulation is small

:- VA is large - small

$$\frac{V_{out}}{V_{in}} = \frac{g_m}{g_m + \frac{1}{r_o} + \frac{1}{R_F}} \left(\frac{R_F}{R_F + R_S} \right)$$

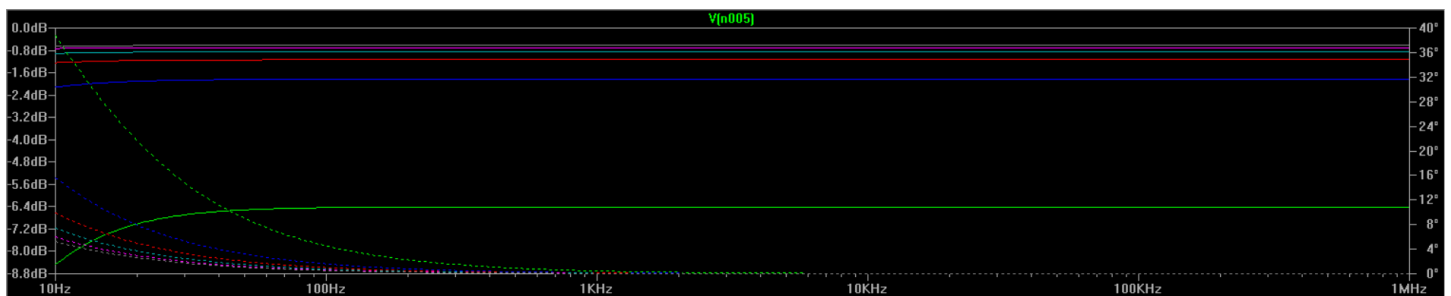
$$\left[\frac{V_{out}}{V_{in}} = \frac{g_m R_F}{g_m R_F + 1 + \frac{R_F}{R_S}} \right]$$

$$\frac{V_{out}}{V_{in}} = \frac{g_m R_F}{g_m R_F + 1 + \frac{R_F}{R_S}}$$

$$\frac{V_{out}}{V_{in}} = \frac{g_m R_F}{g_m R_F + 1 + \frac{R_F}{R_S}}$$

we know that

Ans 2c



```
.INCLUDE tsmc018.lib
```

```
.step param RL 1K 20K 4K
```

